



Family Name

Given Name

Student No.

Signature

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

MID-TERM EXAMINATION

Term 3, 2019

ELEC1111

Electrical and Telecommunications Engineering

TIME ALLOWED:	75 min
TOTAL MARKS:	100
TOTAL NUMBER OF QUESTIONS:	5

THIS EXAM CONTRIBUTES 25% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 5 minutes.

This paper contains 5 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [15 marks]

a. (9 marks) For the circuit shown in Figure 1,

- (7 marks) Calculate the equivalent resistance R_{eq} as seen from terminals $a-b$.
- (2 marks) Find current i using the result of part (i).

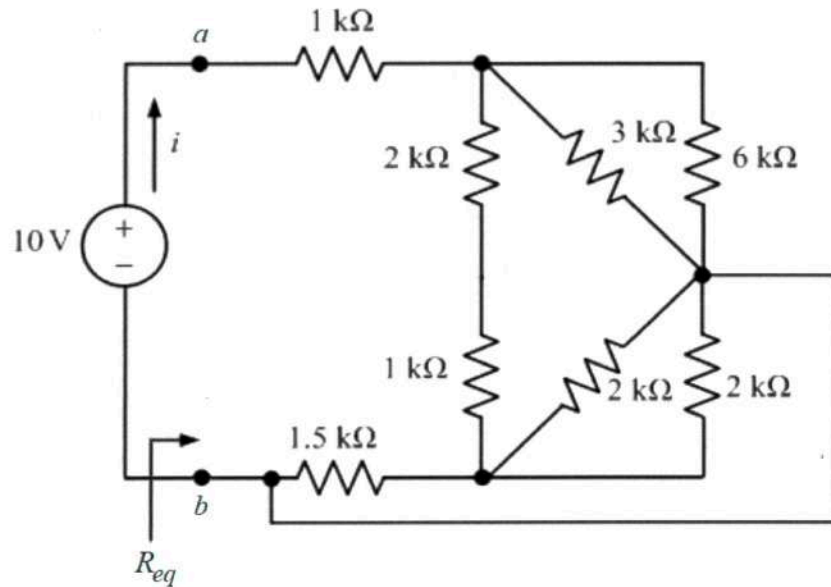
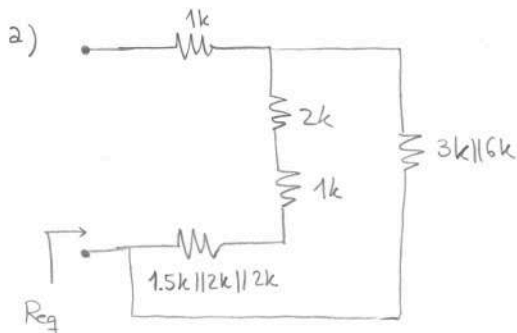


Figure 1



$$\begin{aligned}
 R_{eq} &= \left[\left[(1.5 \parallel 2 \parallel 2) + 1 + 2 \right] \parallel (3 \parallel 6) \right] + 1 \text{ k}\Omega = \\
 &= \left[(0.6 + 1 + 2) \parallel 2 \right] + 1 = \\
 &= 3.6 \parallel 2 + 1 = 1.286 + 1 = \boxed{2.286 \text{ k}\Omega}
 \end{aligned}$$

b)

$$i = \frac{10\text{V}}{2.286 \text{ k}\Omega} = 4.37 \text{ mA}$$

- b. **(6 marks)** An electric pencil sharpener rated 240 mW, 6 V is connected to a 9 V battery as shown in Figure 2. Calculate the value of the series-dropping resistor, R_x , needed to power the sharpener.

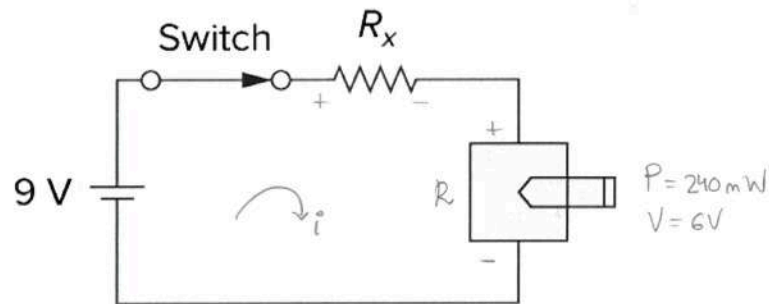


Figure 2

First, calculate sharpener's current and resistance:

$$P = V \times i \Rightarrow i = \frac{P}{V} = \frac{240 \text{ mW}}{6 \text{ V}} = 40 \text{ mA}$$

$$R = \frac{V}{i} = \frac{6 \text{ V}}{40 \text{ mA}} = 150 \Omega$$

Then, calculate R_x :

$$V_{R_x} = 9 \text{ V} - 6 \text{ V} = 3 \text{ V} \Rightarrow \boxed{R_x = \frac{V_{R_x}}{i} = \frac{3 \text{ V}}{40 \text{ mA}} = 75 \Omega}$$

QUESTION 2 [25 marks]

a. (13 marks) For the circuit shown in Figure 3,

- (8 marks) Apply nodal analysis and write down the nodal equations using the labels provided. Note: Simplify the equations, but DO NOT solve them.
- (5 marks) Given the values of nodal voltages as $v_1 = 100\text{ V}$, $v_2 = 550\text{ V}$ and $v_3 = 400\text{ V}$, find the power in the 300 V voltage source.

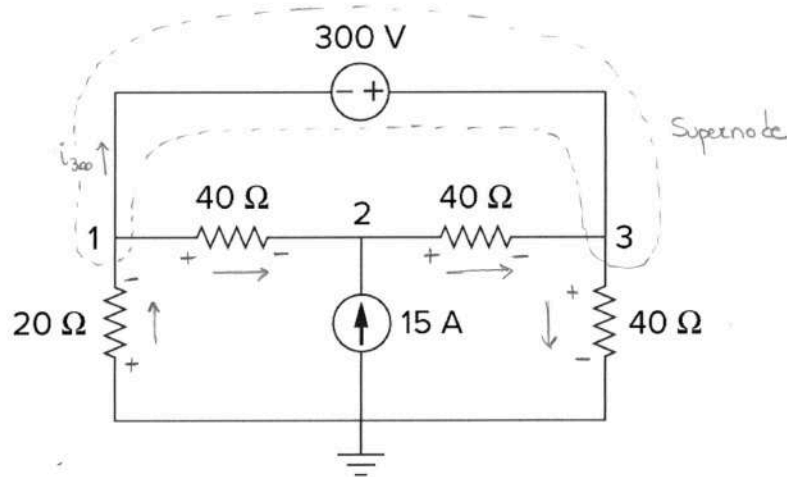


Figure 3

i)

* KCL @ supernode :

$$\frac{0 - V_1}{20} - \frac{V_1 - V_2}{40} + \frac{V_2 - V_3}{40} - \frac{V_3}{40} = 0$$

$$-\left(\frac{1}{20} + \frac{1}{40}\right)V_1 + \left(\frac{1}{40} + \frac{1}{40}\right)V_2 - \left(\frac{1}{40} + \frac{1}{40}\right)V_3 = 0$$

$$-0.075V_1 + 0.05V_2 - 0.05V_3 = 0$$

* KCL @ node 2 :

$$\frac{V_1 - V_2}{40} + 15 - \frac{V_2 - V_3}{40} = 0$$

$$\frac{1}{40}V_1 - \left(\frac{1}{40} + \frac{1}{40}\right)V_2 + \frac{1}{40}V_3 = -15$$

$$0.025V_1 - 0.05V_2 + 0.025V_3 = -15$$

$$\Rightarrow \begin{cases} -0.075V_1 + 0.05V_2 - 0.05V_3 = 0 \\ 0.025V_1 - 0.05V_2 + 0.025V_3 = -15 \\ -V_1 + V_3 = 300 \end{cases}$$

* Constraint equation :

$$V_3 - V_1 = 300$$

ii)

$$\frac{0 - V_1}{20} = i_{300} + \frac{V_1 - V_2}{40} \Rightarrow i_{300} = \frac{-100}{20} - \frac{100 - 550}{40} = -5 + \frac{450}{40} = 6.25\text{ A}$$

$$\boxed{P_{300} = -300 \times i_{300} = -300 \times 6.25 = -1875\text{ W}}$$

- b. (12 marks) Bipolar transistors can serve as amplifiers, producing both current gain and voltage gain. The circuitry in the dashed box models a bipolar junction transistor and the whole circuit in Figure 4 models a transistor amplifier.

For the transistor amplifier shown in Figure 4, apply mesh analysis and write down the mesh equations using the labels provided. Note: Simplify the equations, but DO NOT solve them.

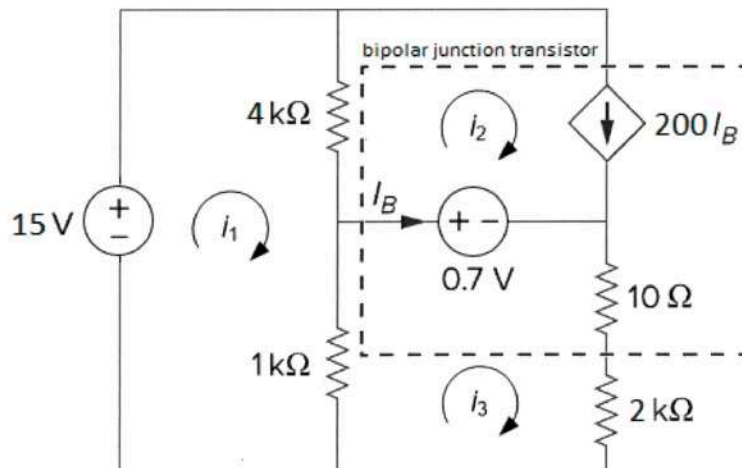
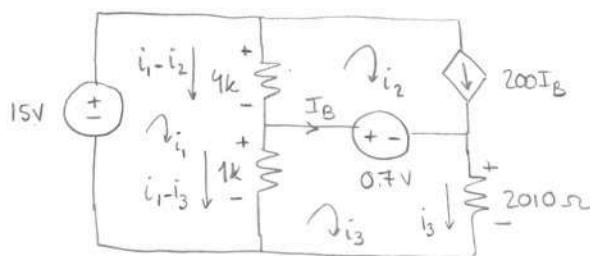


Figure 4



* KVL @ mesh 1: $-15 + 4k(i_1 - i_2) + 1k(i_1 - i_3) = 0$

* $i_2 = 200 I_B$, where $I_B = i_3 - i_2 \Rightarrow i_2 = 200(i_3 - i_2)$

* KVL @ mesh 3: $-1k(i_1 - i_3) + 0.7 + 2010 i_3 = 0$

Let's simplify:

$$\begin{cases} 5000 i_1 - 4000 i_2 - 1000 i_3 = 15 \\ 201 i_2 - 200 i_3 = 0 \\ -1000 i_1 + 3010 i_3 = -0.7 \end{cases}$$

QUESTION 3 [15 marks]

For the circuit shown in Figure 5, use a succession of source transformations to find the voltage v_0 .

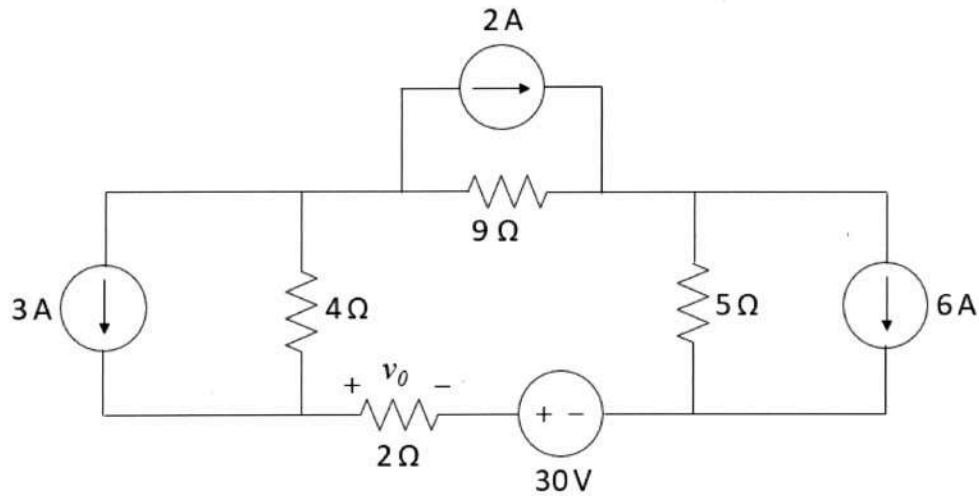
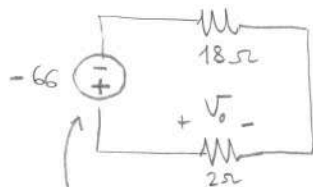
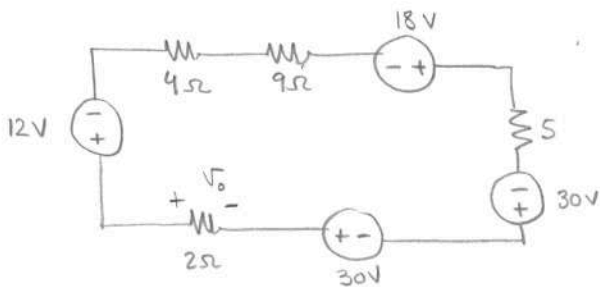


Figure 5



$$V_s = 12 - 18 - 30 - 30 = -66$$

$$\boxed{V_0 = -66 \times \frac{2}{18+2} = -6.6 \text{ V}}$$

QUESTION 4 [20 marks]

The circuit shown in Figure 6 is being used to power a hot water heater. The heater can be modelled as a single resistor connected to terminals $a-b$.

- (9 marks)** Calculate the Thevenin voltage at terminals $a-b$.
- (9 marks)** Calculate the heater resistance that will ensure the maximum transfer of power from the circuit to the heater.
- (2 marks)** Find the maximum power that can be delivered to the heater.

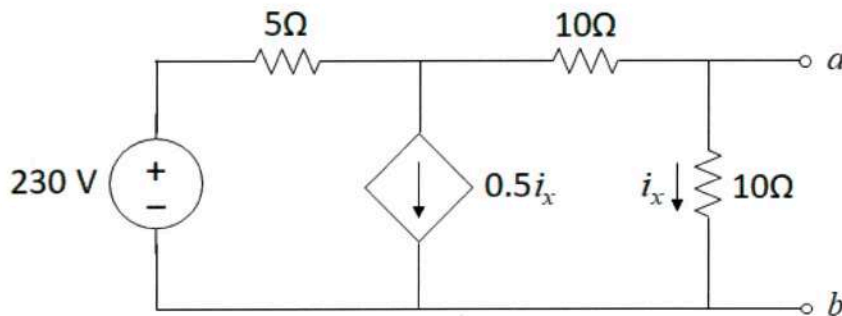
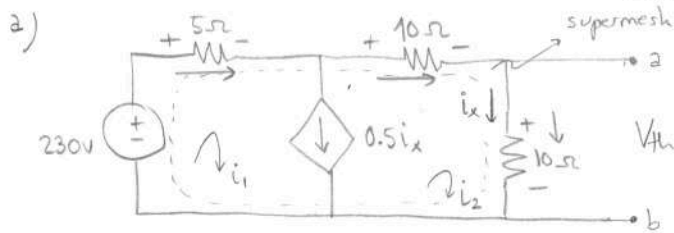


Figure 6



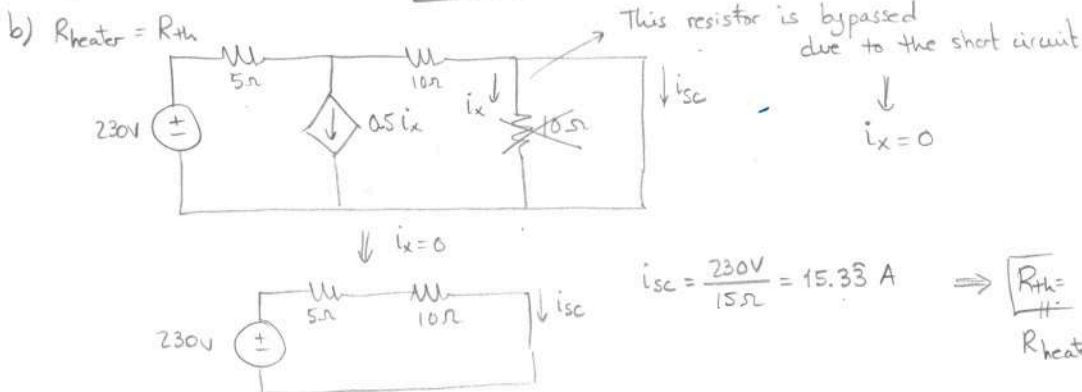
Solution using mesh analysis:

$$\text{KVL @ supermesh: } -230 + 5i_1 + 10i_2 + 10i_2 = 0 \Rightarrow 5i_1 + 20i_2 = 230$$

$$\text{constraint eq: } i_1 - i_2 = 0.5i_x, \text{ where } i_x = i_2 \Rightarrow i_1 - 1.5i_2 = 0$$

$$\begin{aligned} 5i_1 + 20i_2 &= 230 \\ i_1 - 1.5i_2 &= 0 \end{aligned} \Rightarrow \begin{aligned} 5i_1 + 20i_2 &= 230 \\ -5i_1 + 7.5i_2 &= 0 \end{aligned} \xrightarrow{+(-5)} \begin{aligned} 27.5i_2 &= 230 \Rightarrow i_2 = \frac{230}{27.5} = 8.363 \text{ A} \end{aligned}$$

$$\boxed{V_{th} = V_{ab} = V_{10\Omega} = 10i_2 = 83.63 \text{ V}}$$



$$c) \boxed{P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{83.63^2}{4 \times 5.45} = 320.82 \text{ W}}$$

QUESTION 5 [25 marks]

- a. (10 marks) Find the voltage across the capacitor, v_C , and the current flowing through the inductor, i_L , in the circuit of Figure 7 under DC steady state conditions.

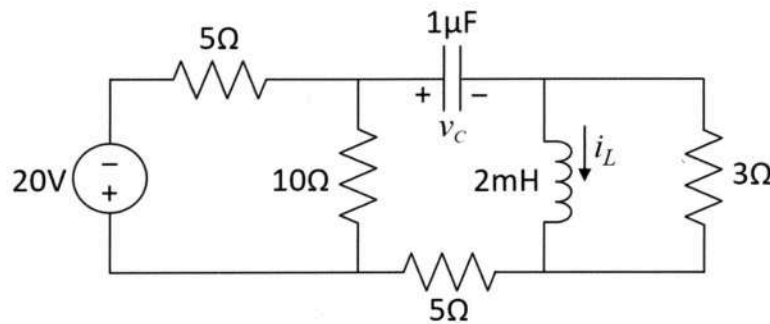
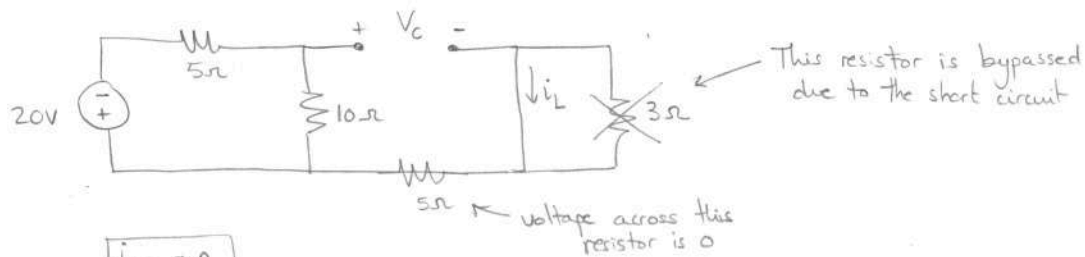


Figure 7

Under DC conditions (steady state) $\left\{ \begin{array}{l} C \rightarrow \text{open circuit} \\ L \rightarrow \text{short circuit} \end{array} \right.$



$$i_L = 0 \text{ A}$$

$$V_C = -V_{10\Omega} = -20 \frac{10}{5+10} = -\frac{200}{15} = -13.33 \text{ V}$$

b. (15 marks) The circuit in Figure 8 is designed to operate an alarm which activates when the current through it exceeds 0.1 mA.

- Calculate the voltage across the capacitor for $t > 0$ s. Assume that the capacitor was initially discharged.
- How long does it take for the alarm to activate for the first time after the switch is closed?

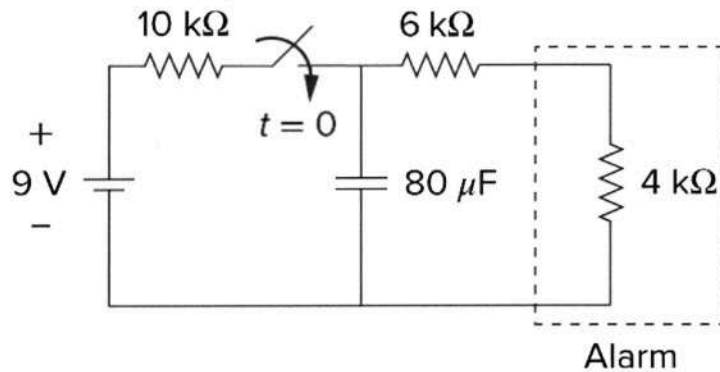
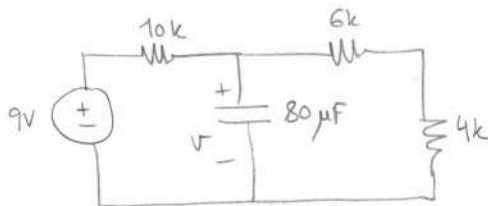


Figure 8

i)
 $t > 0$



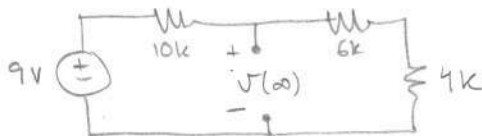
Step response:

$$v(t) = v(\infty) + (v(0) - v(\infty))e^{-t/\tau} \quad (1)$$

$$\text{where } \tau = R_{th}C$$

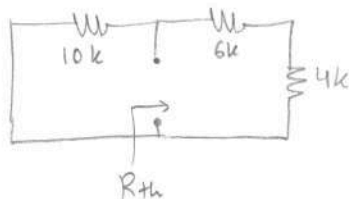
* Calculate $v(0) = v(0^-) = v(0^+)$ for $t < 0 \Rightarrow v(0) = 0V$ (capacitor initially discharged)

* Calculate $v(\infty)$ for $t > 0$ (capacitor behaves as open circuit):



$$v(\infty) = V_{6k+4k} = 4.5V \quad (\text{voltage splits equally})$$

* Calculate $\tau = R_{th}C$ for $t > 0$



$$R_{th} = 10k \parallel 10k = 5k$$

$$\tau = R_{th}C = 5k \times 80\mu F = 0.4s$$

We substitute in (1): $v(t) = 4.5 - 4.5e^{-t/0.4} = 4.5(1 - e^{-t/0.4})V$ for $t > 0$

ii) $i_{4k} \geq 0.1mA$ for alarm to activate

$$V_{4k} = v(t) \frac{4k}{6k+4k} = 0.4v(t) = 1.8(1 - e^{-t/0.4})$$

voltage divider

$$i_{4k} = \frac{V_{4k}}{4k} = 0.45(1 - e^{-t/0.4})mA = 0.1mA$$

$$-e^{-t/0.4} = \frac{0.1}{0.45} - 1 = -0.777$$

$$\frac{-t}{0.4} = \ln(0.777) = -0.251$$

$$t = 0.1s$$